Article Summary

GlioMap: MATLAB-based GUI for MRI Brain Tumor Segmentation and Feature Extraction

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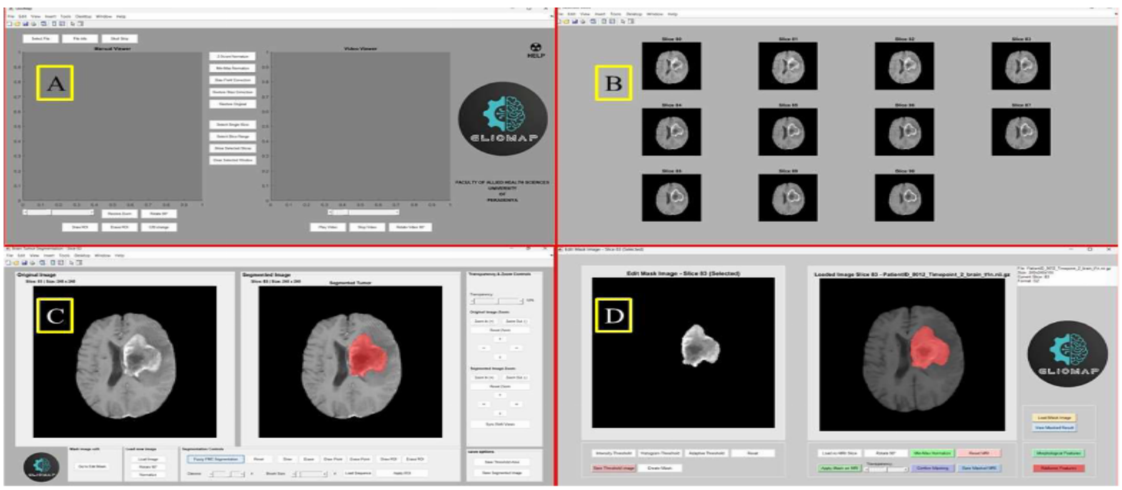
SoftwareX 34 (2026), 102498

1. Purpose and Significance

GlioMap is an open-source MATLAB application designed to make MRI brain-tumor segmentation and quantitative feature extraction accessible through a graphical user interface. It combines preprocessing, skull stripping, semi-automatic or manual segmentation, tumor-mask management, and radiomic and morphological feature calculation in one workflow. The central motivation is to reduce the programming and software-integration burden faced by students, clinicians, and researchers while producing standardized quantitative imaging outputs.

A particularly important design goal is **reproducibility**. GlioMap reports 100 radiomic features and 33 morphological features defined according to the Image Biomarker Standardization Initiative (IBSI). Tumor masks and feature spreadsheets can be exported for statistical analysis, machine-learning pipelines, or other neuro-oncology research workflows.



:. 1. The four main windows of GlioMap: (A) Main window; (B) Selected slice window; (C) Segmentation window; (D) Edit mask window. These modules repres
sequential workflow of the software, beginning with image loading and preprocessing, progressing through tumor segmentation, and concluding with m
ting and feature extraction. Example images are obtained from The Cancer Imaging Archive (TCIA) [12].
lditional information related to software architecture is provided in supplementary material _ S3)

Article Fig. 1 (p. 2). GlioMap's four principal GUI windows: main window, selected-slice window, segmentation window, and edit-mask window. Together they form the sequential image-to-features workflow.

2. Software Architecture and Inputs

GlioMap v1.0 operates on **2D brain MRI with slice-by-slice visualization**. Multimodal intra-subject sequences such as T1, T2, and FLAIR are expected to be pre-aligned; atlas-based registration is not included in the current release. The program supports numerous medical-image formats, including NIfTI, DICOM, NRRD, MetaImage, Analyze, FreeSurfer, GIPL, VTK, MINC, and conventional image formats.

3. Preprocessing: Normalization, Bias Correction, and Skull Stripping

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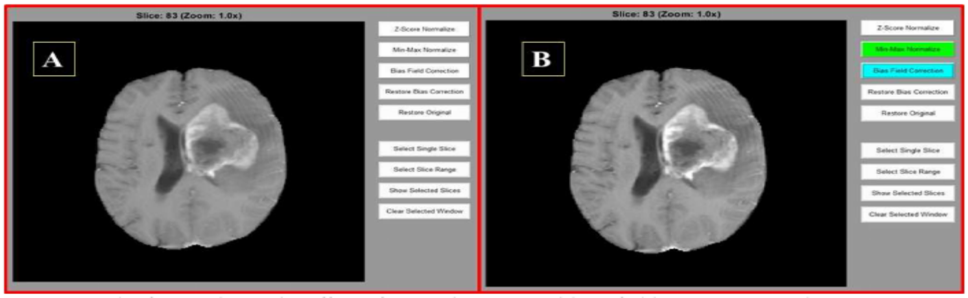
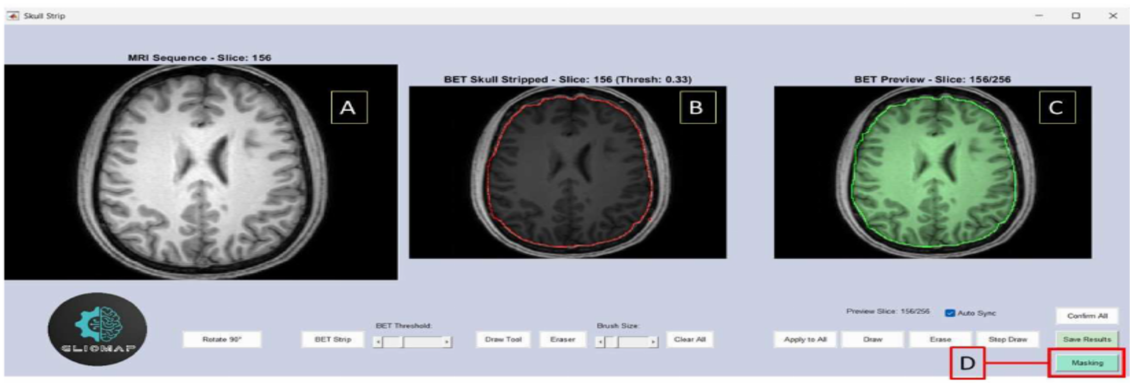


ig. 2. The figure shows the effect of normalization and bias field correction on the image sequence. (A) Before applying the normalization and bias field correctior B) After applying normalization and bias field correction.



ig. 3. Skull stripping sub-window. (A) loaded T1w sequence preview panel, (B) This panel previews the selected slice by the user from the loaded sequence, and th ser can apply BET and can adjust the threshold value, (C) This panel previews the application of the adjusted threshold value to the whole sequence. (D) Maskin ption for other MRI sequences. (T1w MRI data obtained from the OpenNeuro dataset [13].

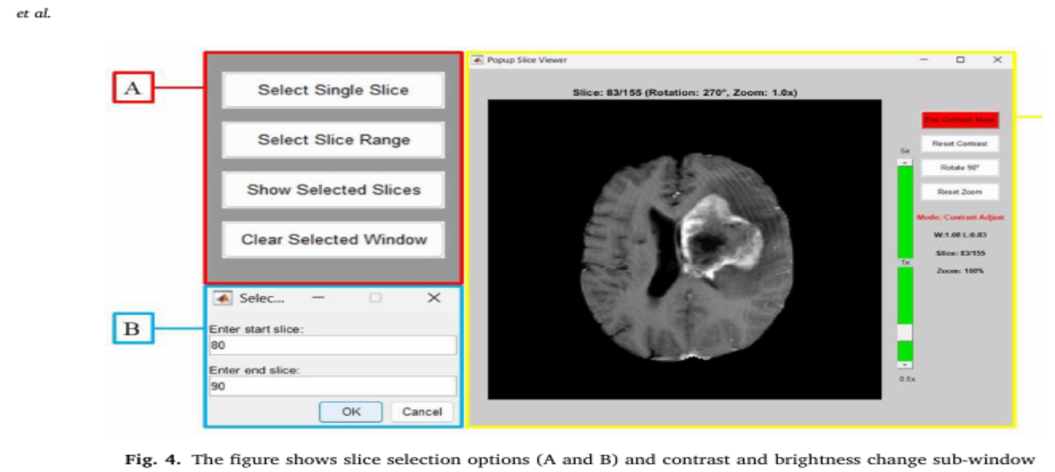
Article Figs. 2-3 (p. 3). Top: image appearance before and after normalization/bias-field correction. Bottom: the skull-stripping interface, including threshold adjustment, preview, and masking.

Preprocessing includes **z-score normalization**, **min-max normalization**, contrast/brightness adjustment, bias-field correction, and skull stripping. The bias-field procedure is an adaptation inspired by the N4 multiplicative-bias framework, but it omits the full iterative B-spline estimation used in N4 ITK and instead applies a faster slice-wise correction suitable for integration into the MATLAB GUI.

For skull stripping, GlioMap implements a simplified Brain Extraction Tool (BET)-style pipeline using Gaussian smoothing, fixed intensity thresholding, connected-component selection, hole filling, and other morphological operations. This simplification is intended to reduce algorithmic complexity and works best when T1-weighted MRI provides sufficient brain/non-brain intensity contrast.

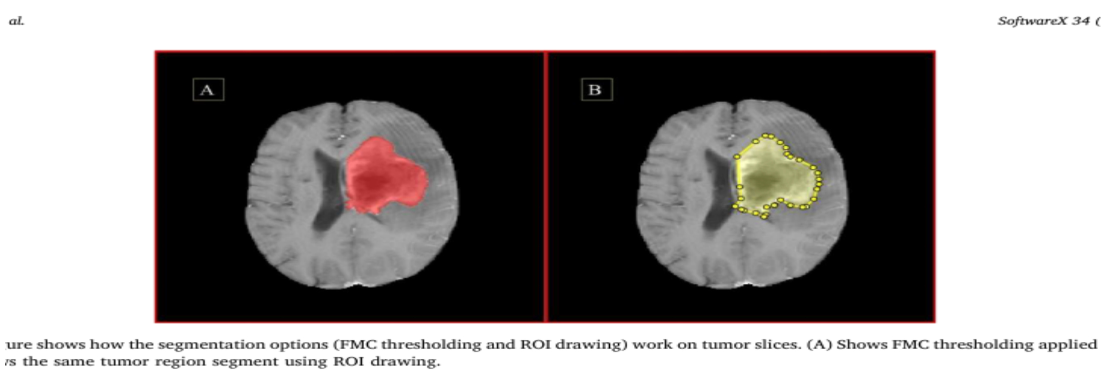
The authors informally tested the simplified skull stripping method on 10 randomly selected T1-weighted OpenNeuro cases. The masks were visually judged to preserve tumor regions and support downstream processing. Importantly, the paper explicitly characterizes this as a **small feasibility check rather than a formal benchmark** and does not position GlioMap as a replacement for more robust skull-stripping systems.

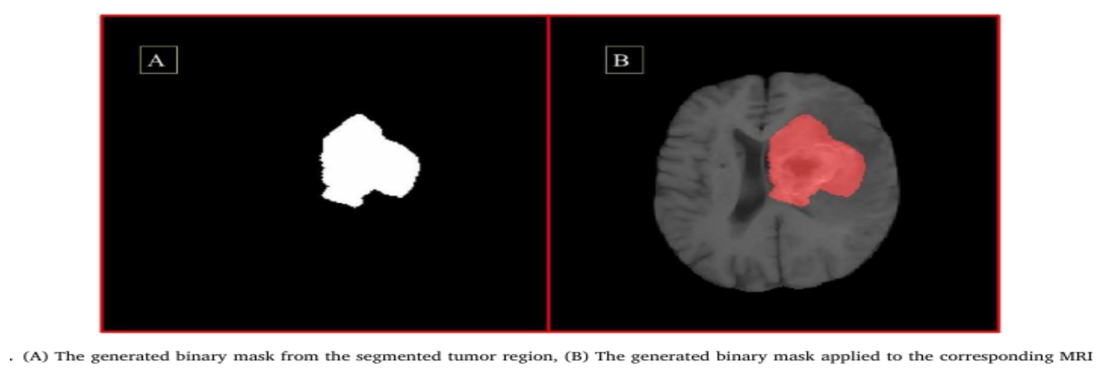
4. Slice Selection and Tumor Segmentation



Article Fig. 4 (p. 4). Slice-selection controls and the contrast/brightness adjustment window used to prepare tumor-containing slices for segmentation.

Users can select an individual tumor slice or a range of slices and adjust contrast and brightness before segmentation. GlioMap provides two segmentation routes: **FMC thresholding** for semi-automatic segmentation and **manual ROI drawing**. In the FMC formulation, fuzzy cluster membership identifies a high-intensity tumor candidate, adaptive thresholding is applied, and morphological closing, small-area removal, hole filling, and final cleanup produce the tumor mask.





Article Figs. 5-6 (p. 6). Fig. 5 contrasts FMC thresholding with manual ROI drawing. Fig. 6 shows the resulting binary tumor mask and the mask overlaid/applied to the MRI slice.

After segmentation, users can refine the result in the Edit Mask Window. The software supports binary-mask generation, morphological cleanup, connected-component operations, saving the final mask, and applying a generated or previously saved mask to the tumor image. This hybrid semi-automatic/manual design is intended to retain user control when tumor boundaries are ambiguous.

5. Radiomic and Morphological Feature Extraction

Once a tumor ROI has been defined, GlioMap calculates **100 radiomic features** and **33 morphological features**. Radiomics quantify intensity and texture characteristics relevant to tumor heterogeneity, while morphological descriptors quantify tumor geometry. Examples discussed in the paper include volume, surface area, compactness, sphericity, elongation, and bounding-box dimensions.

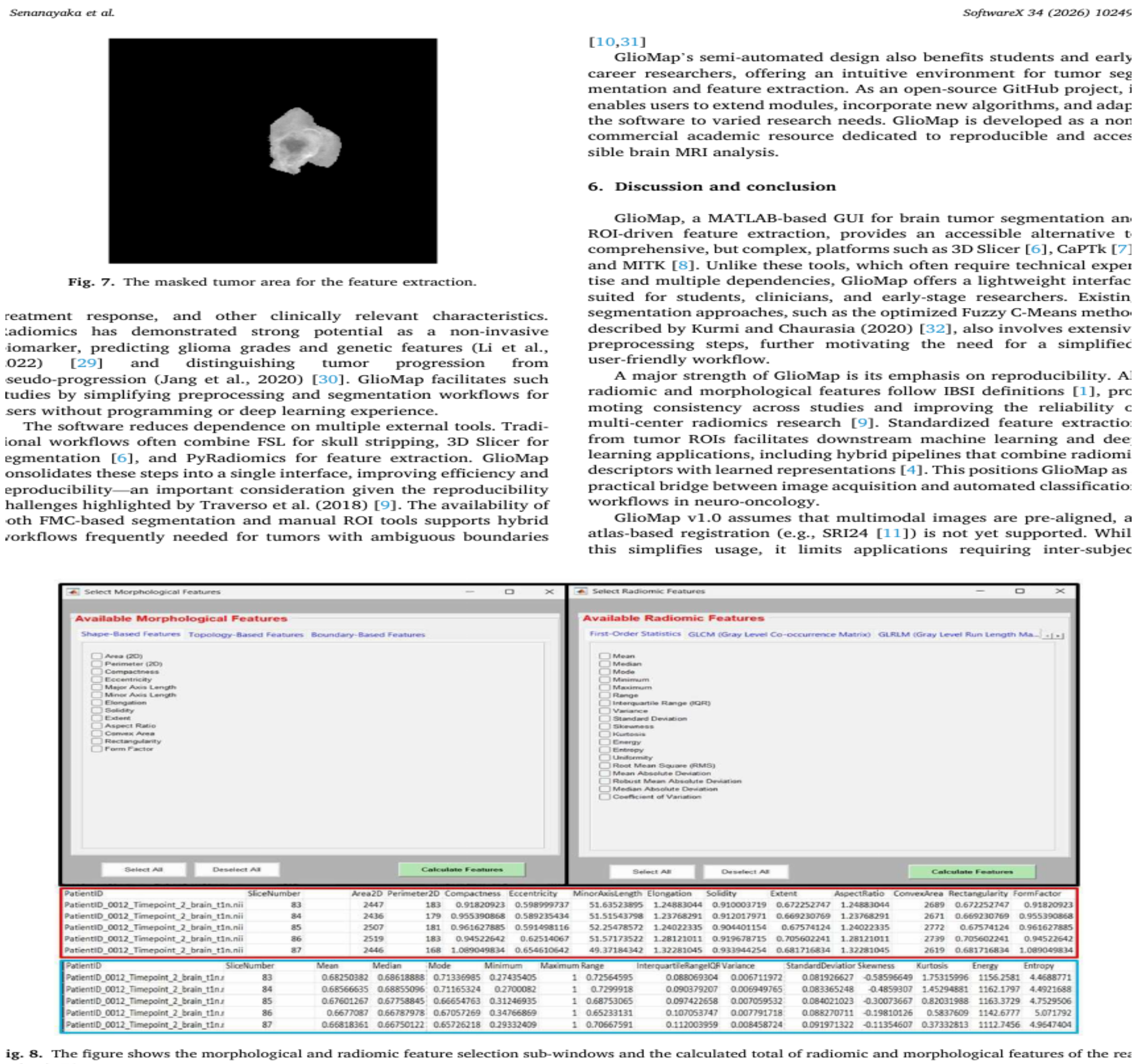


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Fig. 7. The masked tumor area for the feature extraction.

reatment response, and other clinically relevant characteristics. adiomics has demonstrated strong potential as a non-invasive iomarker, predicting glioma grades and genetic features (Li et al., 022) [29] and distinguishing tumor progression from seudo-progression (Jang et al., 2020) [30]. GlioMap facilitates such tudies by simplifying preprocessing and segmentation workflows for sers without programming or deep learning experience.

The software reduces dependence on multiple external tools. Tradi-ional workflows often combine FSL for skull stripping, 3D Slicer for egmentation [6], and PyRadiomics for feature extraction. GlioMap onsolidates these steps into a single interface, improving efficiency and eproducibility—an important consideration given the reproducibility hallenges highlighted by Traverso et al. (2018) [9]. The availability of oth FMC-based segmentation and manual ROI tools supports hybrid vorkflows frequently needed for tumors with ambiguous boundaries [10,31]

GlioMap's semi-automated design also benefits students and early career researchers, offering an intuitive environment for tumor seg mentation and feature extraction. As an open-source GitHub project, i enables users to extend modules, incorporate new algorithms, and adap the software to varied research needs. GlioMap is developed as a non commercial academic resource dedicated to reproducible and acces sible brain MRI analysis.

6. Discussion and conclusion

GlioMap, a MATLAB-based GUI for brain tumor segmentation an ROI-driven feature extraction, provides an accessible alternative t comprehensive, but complex, platforms such as 3D Slicer [6], CaPTk [7] and MITK [8]. Unlike these tools, which often require technical exper tise and multiple dependencies, GlioMap offers a lightweight interfac suited for students, clinicians, and early-stage researchers. Existin segmentation approaches, such as the optimized Fuzzy C-Means metho described by Kurmi and Chaurasia (2020) [32], also involves extensiv preprocessing steps, further motivating the need for a simplified user-friendly workflow.

A major strength of GlioMap is its emphasis on reproducibility. A radiomic and morphological features follow IBSI definitions [1], prc moting consistency across studies and improving the reliability c multi-center radiomics research [9]. Standardized feature extractio from tumor ROIs facilitates downstream machine learning and dee learning applications, including hybrid pipelines that combine radiomi descriptors with learned representations [4]. This positions GlioMap as practical bridge between image acquisition and automated classificatio workflows in neuro-oncology.

GlioMap v1.0 assumes that multimodal images are pre-aligned, a atlas-based registration (e.g., SRI24 [11]) is not yet supported. Whil this simplifies usage, it limits applications requiring inter-subjec

PatientID	SliceNumber	Area2D	Perimeter2D	Compactness	Eccentricity	MinorAxisLength	Elongation	Solidity	Extent	AspectRatio	ConvexArea	Rectangularity	FormFactor
PatientID_0012_Timepoint_2_brain_t1n.nii	83	2447	183	0.91820923	0.598999737	51.63523895	1.24883044	0.910003719	0.672252747	1.24883044	2689	0.672252747	0.91820923
PatientID_0012_Timepoint_2_brain_t1n.nii	84	2436	179	0.955390868	0.589235434	51.51543798	1.23768291	0.912017971	0.669230769	1.23768291	2671	0.669230769	0.955390868
PatientID_0012_Timepoint_2_brain_t1n.nii	85	2507	181	0.961627885	0.591498116	52.25478572	1.24022335	0.904401154	0.67574124	1.24022335	2772	0.67574124	0.961627885
PatientID_0012_Timepoint_2_brain_t1n.nii	86	2519	183	0.94522642	0.62514067	51.57173522	1.28121011	0.919678715	0.705602241	1.28121011	2739	0.705602241	0.94522642
PatientID_0012_Timepoint_2_brain_t1n.nii	87	2446	168	1.089049834	0.654610642	49.37184342	1.32281045	0.933944254	0.681716834	1.32281045	2619	0.681716834	1.089049834

PatientID	SliceNumber	Mean	Median	Mode	Minimum	Maximum	Range	InterquartileRangeIQR	Variance	StandardDeviation	Skewness	Kurtosis	Energy	Entropy
PatientID_0012_Timepoint_2_brain_t1n.r	83	0.68250382	0.68618888	0.71336985	0.27435405	1	0.72564595	0.088069304	0.006711972	0.081926627	-0.58596649	1.75315996	1156.2581	4.4688771
PatientID_0012_Timepoint_2_brain_t1n.r	84	0.68566635	0.68855096	0.71165324	0.2700082	1	0.7299918	0.090379207	0.006949765	0.083365248	-0.4859307	1.45294881	1162.1797	4.4921688
PatientID_0012_Timepoint_2_brain_t1n.r	85	0.67601267	0.67758845	0.66654763	0.31246935	1	0.68753065	0.097422658	0.007059532	0.084021023	-0.30073667	0.82031988	1163.3729	4.7529506
PatientID_0012_Timepoint_2_brain_t1n.r	86	0.6677087	0.66787978	0.67057269	0.34766869	1	0.65233131	0.107053747	0.007791718	0.088270711	-0.19810126	0.5837609	1142.6777	5.071792
PatientID_0012_Timepoint_2_brain_t1n.r	87	0.66818361	0.66750122	0.65726218	0.29332409	1	0.70667591	0.112003959	0.008458724	0.091971322	-0.11354607	0.37332813	1112.7456	4.9647404

ig. 8. The figure shows the morphological and radiomic feature selection sub-windows and the calculated total of radiomic and morphological features of the rea atient's T1w sequence tumor core area.

Article Figs. 7-8 (p. 7). The masked tumor ROI used for feature extraction and the GUI panels for selecting morphological/radiomic features, with calculated feature values displayed in tabular form.

The paper emphasizes IBSI compliance because standardized feature definitions are important for repeatability, reproducibility, multi-center radiomics studies, and downstream machine-learning models. Extracted features can be saved as spreadsheets, while preprocessed images, skull-stripped images, and tumor masks can also be exported.

6. End-to-End Workflow

The illustrative workflow presented in the article can be summarized as: **(1)** load multimodal MRI; **(2)** normalize and correct bias field; **(3)** perform skull stripping; **(4)** select tumor-containing slices and adjust visualization; **(5)** segment using FMC or manual ROI drawing; **(6)** edit and save the tumor mask; **(7)** calculate radiomic and morphological features; and **(8)** export images, masks, and feature files for analysis.

The authors also report a small usability exercise with five undergraduate students. The same preprocessing, segmentation, and feature-extraction workflow took approximately **20 minutes in GlioMap versus 40-50 minutes in CaPTk**. This is presented as evidence of usability for teaching and rapid prototyping, although the very small sample means it should not be interpreted as a broad comparative performance study.

7. Limitations

GlioMap v1.0 has several important limitations. It currently performs **2D, slice-wise processing**, so manual or semi-automatic work can become time-consuming on large datasets. Multimodal scans must already be aligned because atlas-based registration is absent. The current tumor-core workflow also relies on contrast-enhanced T1-weighted MRI. In addition, the simplified skull-stripping validation is exploratory rather than a formal quantitative benchmark.

These constraints are central to interpreting the software correctly: GlioMap is presently a lightweight research, teaching, and prototyping environment rather than a complete automated clinical segmentation system.

8. Planned Development

The authors propose future additions including **full 3D processing**, automated machine-learning and deep-learning models for slice selection, segmentation, skull stripping, and feature computation, **atlas-based registration** such as SRI24, and a **contrast-free tumor-core prediction module**. These upgrades would move GlioMap from its current ROI-driven 2D workflow toward more automated and comprehensive quantitative neuro-oncology analysis.

9. Overall Takeaway

GlioMap's main contribution is not a new state-of-the-art deep neural network. Instead, it is an integrated and accessible MATLAB environment that links MRI preprocessing, tumor segmentation, mask refinement, and standardized quantitative feature extraction. Its strongest research-oriented feature is the attempt to make IBSI-compliant radiomic and morphological measurements available without requiring users to assemble multiple software packages or write substantial code.

For medical image computing and quantitative neuroradiology, the paper is especially useful as an example of how **segmentation software can bridge image processing and downstream radiomics/AI**: the tumor mask is not merely an endpoint, but the spatial definition from which standardized quantitative biomarkers are computed and exported for later classification, prediction, and clinical-research analysis.

Source: Senanayaka I, Sangavi T, Jayatilake ML, Hevavithana PB, Kulathunga S. *GlioMap: MATLAB-based GUI for MRI Brain Tumor Segmentation and Feature Extraction.* SoftwareX. 2026;34:102498.